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Isolation Forest Anomaly Detection and Long Short-Term Memory Traffic Prediction

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PROJECT OVERVIEW

- Real-time network monitoring using open-source tools
- Docker-based emulated environment in GNS3
- Prometheus & Grafana for metric collection and visualization
- AI modules for anomaly detection and traffic forecasting
- Full automation and real traffic generation via containers

WHY THIS PROJECT?

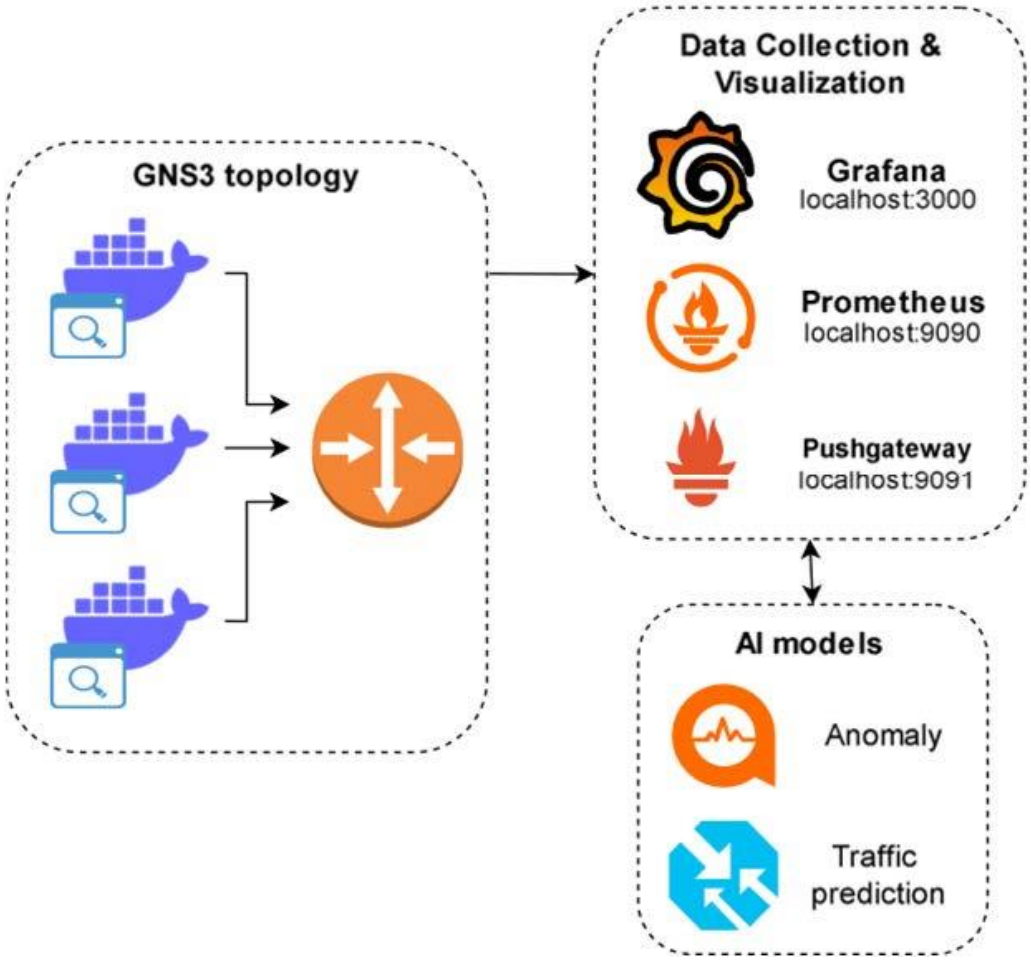
- Traditional monitoring systems:
  - often rely on static thresholds and reactive alerts
  - lack prediction capabilities and real anomaly detection
- Full enterprise legacy tools: Splunk, Cisco DNA Center. Disadvantages:
  - closed-source
  - expensive
  - complex for academic use
- Proposed solution: entirely from scratch, based on:
  - GNS3
  - Docker containers
  - Prometheus and Grafana
  - Python + ML models: Isolation Forest and Long Short-Term Memory Traffic
- Goals:
  - scalable, modular, and intelligent monitoring system
  - easy to deploy and to understand

SYSTEM ARCHITECTURE

GNS3 Topology:
- containers that generate traffic and send it through a router, enabling Prometheus to collect the metrics

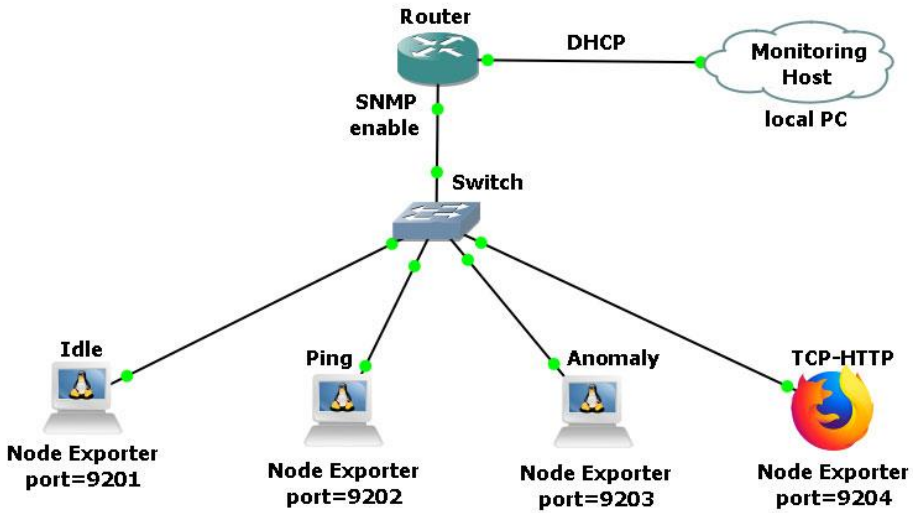
Data Collection & Visualization:
-Prometheus collects relevant metrics, which are visualized in real time using Grafana

AI Models:
- machine learning models that detect anomalies and predict future traffic patterns based on the collected metrics



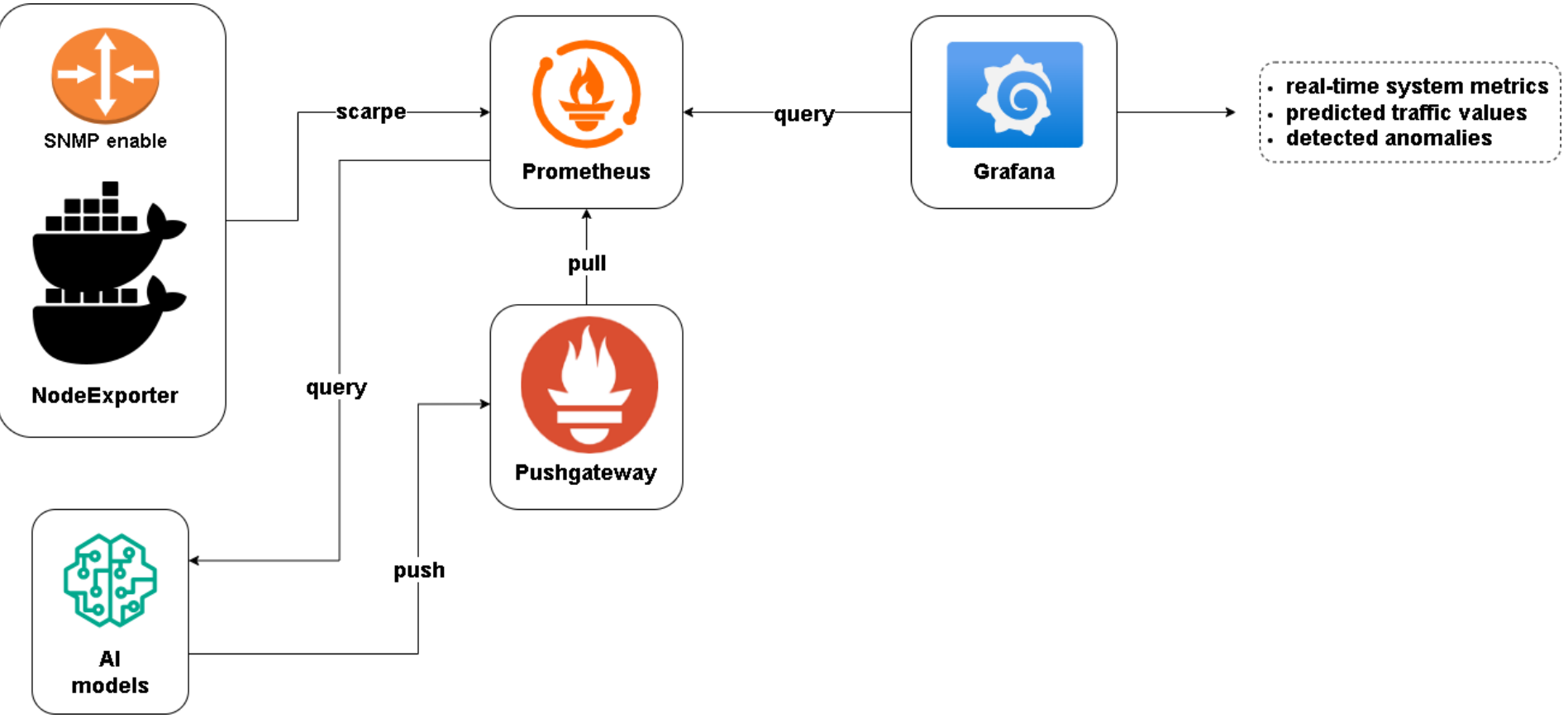
GNS3 VIRTUAL TOPOLOGY

- Docker containers (Alpine-based and Webterm) are connected to a Cisco router
- Each container runs a Node Exporter, which exposes system metrics such as: CPU usage, memory consumption, network I/O (bytes in/out), disk activity
- The Cisco router acts as the central gateway and is connected to the host PC via the GNS3 Cloud node(bridge)
- Port forwarding is configured on the router to allow the local PC to access each container's metrics over unique ports



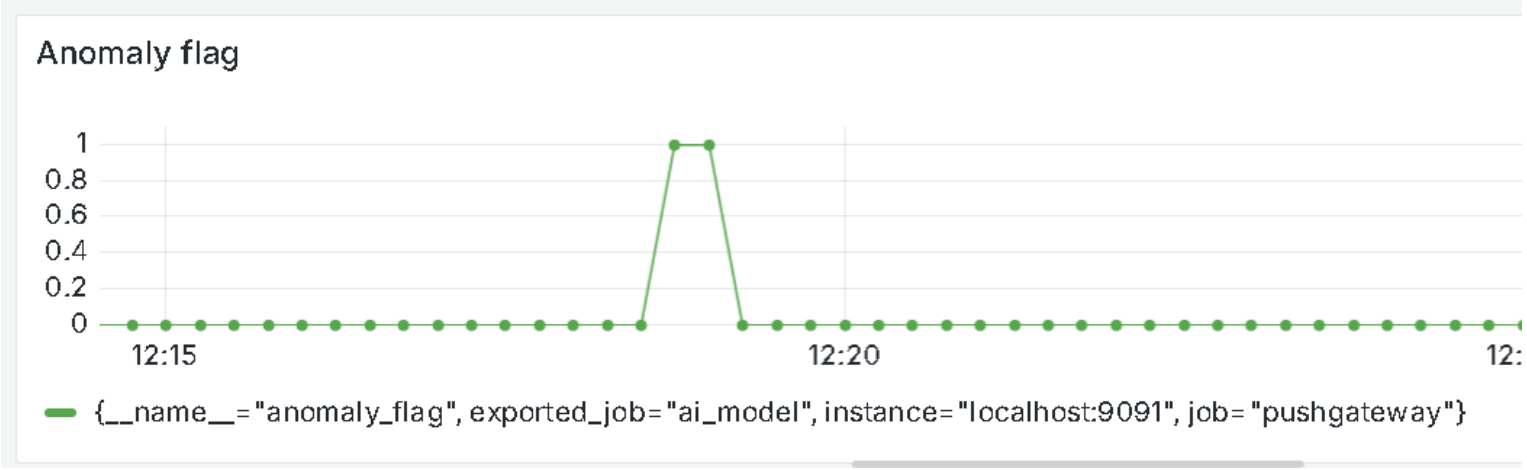
- Traffic is generated from containers to simulate real network activity:
  - idle: no significant traffic
  - ping: sends periodic ICMP requests
  - TCP-HTTP: uses curl/wget to generate HTTP/TCP
  - anomaly: random bursts or scripted heavy load for testing AI detection

METRIC FLOW & PROCESSING



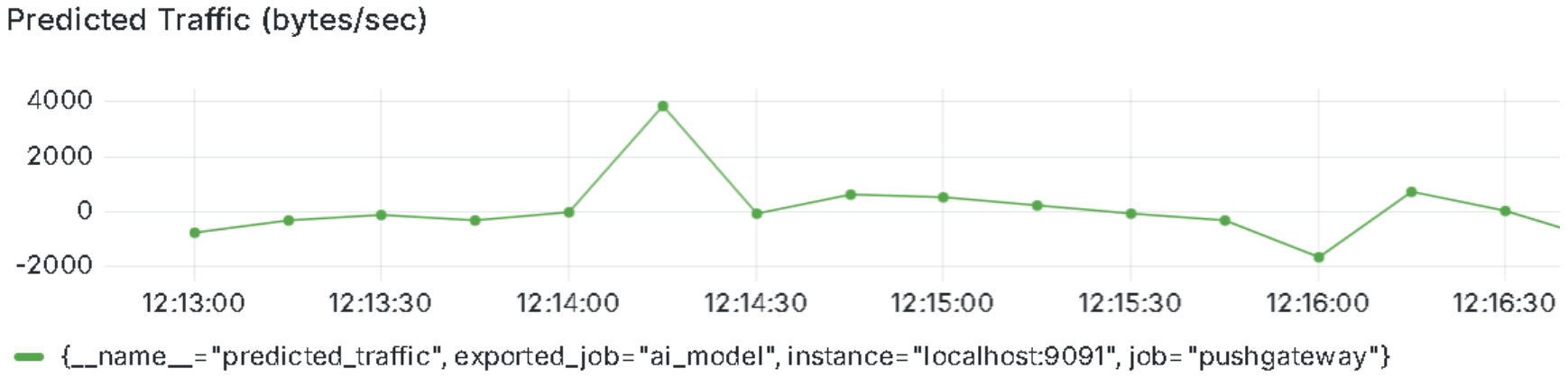
ANOMALY DETECTION (ISOLATION FOREST)

- Model used: Isolation Forest
- Unsupervised algorithm that isolates outliers in metric distributions
- Trained every 60 seconds on the last 10 minutes of data
- Metric used: rate(node_network_receive_bytes_total)
- Output: 0 = normal, 1 = anomaly
- Accuracy observed during test: 87–90%
- Example anomalies: sudden spikes from curl, download bursts



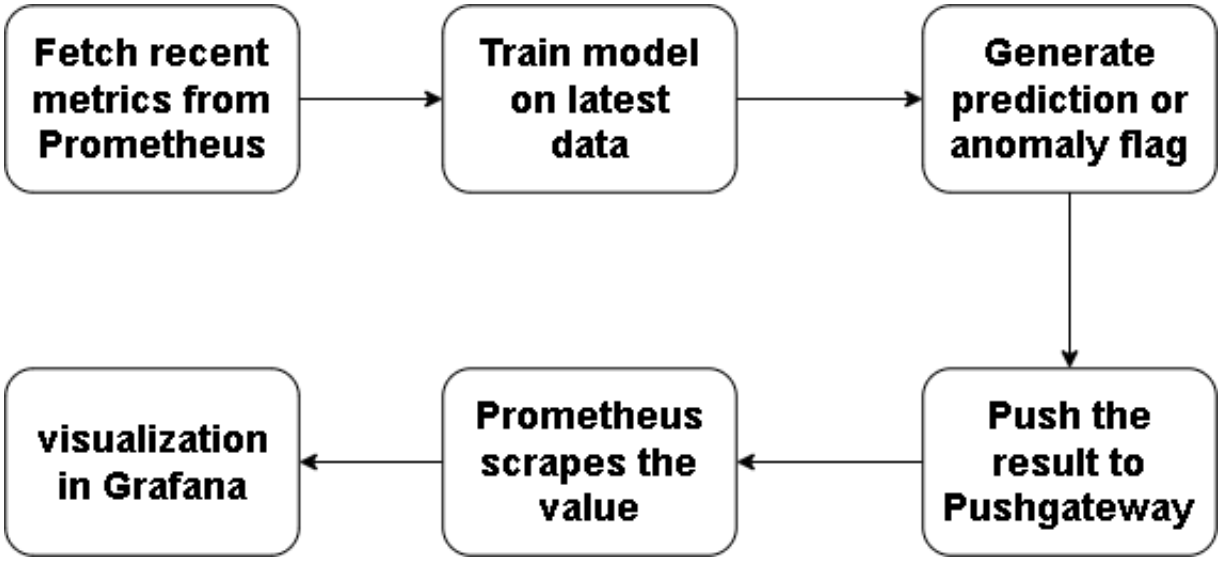
TRAFFIC PREDICTION (LSTM)

- Model used: LSTM (Long Short-Term Memory)
- Designed for time-series forecasting
- Learns patterns from network traffic (bytes/sec over time)
- Windowed training over 10-sample sequences
- Predicts next value and pushes result to Pushgateway
- Accuracy: Mean squared error (MSE) was low (~0.01–0.05) in stable traffic
- Negative outputs indicate model uncertainty or insufficient training data



AI EXECUTION & AUTOMATION

- Python scripts are scheduled to run every 60 seconds using a batch (.bat) script
- Each execution follows this workflow below:



- The process is fully automated and repeats continuously
- No user intervention is required

CONCLUSIONS

- The project demonstrates how AI can enhance traditional monitoring
- Combines:
  - Open-source tools (Prometheus, Grafana, Docker, GNS3)
  - ML models (Isolation Forest, LSTM)
  - Real traffic simulation via containers
- Real-time detection and forecasting is feasible and effective
- Offers flexibility, modularity, and educational value
- Pushgateway enables tight integration between scripts and monitoring tools
- Achieved good prediction reliability and anomaly detection under controlled conditions
- System runs automatically, adapting to recent network data

FUTURE WORK

- Use hybrid models (e.g., Autoencoders + Isolation Forest)
- Save trained models to reduce retraining overhead
- Add more metrics: latency, packet loss, jitter
- Enable feedback loop or alert validation in Grafana
- Expand testbed with multiple routers, subnets, traffic types
- Try supervised ML with labeled attack/normal traffic datasets
- Explore streaming data pipelines (e.g., Kafka + Prometheus)

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